

Faculty of Science & Technology
PES's Modern College of Engineering
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Level 7: First Year M. Tech. (2024 Pattern)

Course Code	CSE08651
Course Name	Skill Development – II
Semester	IV
Experiment No.	8
Title	Statistical Analysis of the Research Project using Python

Problem Statement

To study and analyze research project data using statistical methods in order to derive meaningful conclusions and validate research outcomes.

Objective

To make use of Python-based statistical tools for analyzing research data and interpreting experimental results.

Outcome

CO653.8: Apply statistical analysis techniques to evaluate and interpret research project results.

Software / Hardware Requirements

- Python Programming Environment
- Statistical Libraries (NumPy, SciPy, Pandas)
- Data Visualization Libraries (Matplotlib / Seaborn)
- Personal Computer or Laptop

Theory

Statistical analysis is an essential component of research that enables researchers to summarize, interpret, and draw conclusions from experimental data. It provides quantitative methods for understanding data distributions, relationships among variables, and variability within datasets. Proper statistical analysis ensures that research findings are reliable and scientifically valid.

Python offers a comprehensive ecosystem of libraries for statistical analysis and data science. Libraries such as NumPy and Pandas support numerical computation and data handling, while SciPy provides a wide range of statistical tests and mathematical functions. Visualization libraries further assist in presenting statistical results effectively.

By applying statistical techniques using Python, researchers can validate hypotheses, analyze experimental outcomes, and support data-driven decision-making. The integration of statistical analysis into the research workflow improves result interpretation and strengthens the credibility of research conclusions.

Steps Involved

1. The MNIST dataset was collected and prepared for statistical analysis.
2. Required statistical libraries were imported into the Python environment.
3. Descriptive statistics such as mean and standard deviation were computed.
4. Statistical measures were applied to analyze pixel intensity distributions.
5. Results were visualized and interpreted.

Results

Statistical analysis of the MNIST dataset was successfully performed using Python. Important statistical measures describing digit image characteristics were obtained to support research observations.

Input / Dataset Details

Name of the Dataset: MNIST Handwritten Digits Dataset

Description of the Dataset: The MNIST dataset consists of grayscale images of handwritten digits from 0 to 9. It is commonly used for evaluating statistical analysis, pattern recognition, and machine learning techniques in image-based research.

Dataset Characteristics:

- Image-based dataset
- Labeled supervised dataset
- Balanced digit classes

Subject Area: Machine Learning, Image Processing, Statistical Analysis

Associated Tasks: Digit recognition, statistical feature analysis, image classification

Feature Type: Numerical (pixel intensity values)

Number of Instances: 70,000 images (60,000 training and 10,000 testing samples)

Number of Features: 784 features per instance (28×28 pixel images)

Analysis and Conclusion

Analysis: Statistical evaluation of the MNIST dataset revealed meaningful variations in pixel intensity distributions across digit classes. These measures supported objective analysis of digit patterns.

Conclusion: Statistical analysis using Python provided quantitative insights into the MNIST dataset. Such analysis is essential for validating research results and preparing data for digit recognition and machine learning applications.

References

1. LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P., “Gradient-Based Learning Applied to Document Recognition,” Proceedings of the IEEE.
2. SciPy Community, “SciPy Documentation,” Available: <https://docs.scipy.org>